Powerful Metrics: The Overlooked Factor Behind Softball Pitch Speed

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In softball, pitch speed can change games and often defines a pitcher's success. Athletes and coaches spend countless hours refining their mechanics, aiming for an efficient and consistent motion that maximizes pitch speed. Despite working on traditional pitching performance metrics, the athletes at Planet Fastpitch (PFP), an elite softball training facility in Massachusetts, were not seeing the progress they had hoped for and wanted to further improve their speed. This study aims to determine the best model of measurable attributes to predict the speed of a softball pitch. The primary hypothesis was that strength plays a significant role in pitch speed, in addition to or more than traditional pitch mechanics metrics. Data from 45 athletes at PFP in the spring of 2025 were analyzed using linear regressions. The hypothesis was confirmed, showing that strength is statistically additive to mechanics.

Performance metrics are what pitchers typically seek to optimize as they are building their mechanics. They were obtained through manual video analysis using the Onform app. Power metrics, including grip strength, were collected through standardized assessments of strength. Inhibitors, actions during a pitch that harm speed, were identified by expert coaches reviewing the same videos. All analyses were conducted in Python using the statsmodels.api.ols function on Google Colab.

R^2 was used as a measure of fit, and when computing multiple variable regressions, adjusted R^2 was used because it accounts for the influence of various variables. The equation for adjusted R^2 is:

$$1 - \frac{(1 - R^2)(n - 1)}{(n - k - 1)}$$

Here, R^2 represents the regression, n is the number of data points, and k is the number of independent variables. This formula accounts for degrees of freedom, which penalizes the addition of additional variables. This allows us to determine if the extra variable is truly adding value and not just over-fitting. In other words, R^2 is how well the model explains pitch speed, and adjusted R^2 is used to determine if adding an extra variable actually helps to improve the model.

P values were also used to determine the significance of each variable. P-value is the probability that the null hypothesis is true. In simpler terms, a p-value confirms whether each factor actually impacts speed or if the effect is random. A variable is considered statistically significant if the p-value is less than 0.05 (5%). This was also calculated using the Python statsmodels library.

The following are noteworthy points discovered in the analysis:

- Overall, we have a simple model with only three variables that explains 58% of the variation in the speed. It had an adjusted R^2 of 0.55, and each variable's p-value was < 5%, meaning that it is statistically significant.
- The best single-variable regression between performance metrics and speed was land-to-release (R^2 of 0.255). Several power metrics predict speed with a stronger R^2, most notably the broad jump (R^2 of 0.32), which represents a 45% improvement over performance alone. Additionally, the p-values were all less than 0.0005 for the top power models, indicating that this difference is statistically significant and not random.
- The best two-variable regression was between the grip test and land-to-release, with an adjusted R^2 of 0.47. This is 21% better than performance alone and 14% better than power alone. Further, we see p-values of <0.0005 for each, meaning that power and performance are additive to each other, demonstrating that one is not a better version of the other, and a pitcher needs both.
- The inhibitors were not significant on their own, but they appeared in the best multivariable models, indicating that they are additive once power is controlled.

These findings mean that while performance metrics are essential, a pitcher’s power is an overlooked key factor. Next, the inhibitors and their relationship to speed were observed. Single regressions were performed, and the highest R^2 value was 0.06, indicating that the inhibitors on their own do not statistically significantly impact speed.

We then look at two-variable regressions. In single-variable regressions predicting speed, the highest performance metric produced an R^2 of 0.255, while the highest R^2 for power metrics was 0.32, and the highest R^2 for inhibitors was 0.06. Combining the metrics, a three-variable regression achieved an R^2 of 0.58 and an adjusted R^2 of 0.54, indicating that strength (p-value < 0.0005) is significantly additive to the predictive power of performance metrics. Similarly, inhibitors become additive to multivariable models.

The results support the hypothesis that strength is an overlooked factor in improving speed. While performance metrics are essential for pitchers at PFP who have a solid foundation in performance metrics and mechanics, a key piece for development is to improve their strength. PFP can now share these findings with their pitchers so they know targeted and overall areas for strength improvement. Future work could explore how strength affects hitting and overhand throwing, or research ideas for subjects with various skill levels and experiences.



Teagan Kavan, pitcher for Texas, led her team to win the 2025 Women's College World Series

Three Variable Regression using any Metric or Inhibitor to Predict Speed (Top 10 by Adjusted R^2)

Variable 1	Variable 2	Variable 3	R^2	Adjusted R^2	Variable 1 p Value	Variable 2 p Value	Variable 3 p Value	Number of data points
Grip Test	Land to release	Losing space	0.580	0.545	0.000	0.001	0.027	40
Grip Test	Power Score	Losing space	0.557	0.522	0.001	0.001	0.014	41
Grip Test	Power Score	Land to release	0.551	0.514	0.002	0.049	0.007	41
Grip Test	Land to release	Anchoring	0.551	0.513	0.000	0.001	0.111	40
Grip Test	Sit Up Throw	Land to release	0.541	0.504	0.003	0.078	0.004	41
Grip Test	Seated Chest Pass	Land to release	0.541	0.504	0.001	0.078	0.002	41
Grip Test	Land to release	Marching	0.537	0.498	0.000	0.000	0.231	40
Grip Test	Land to release	Leaning	0.533	0.494	0.000	0.001	0.283	40
Grip Test	Sit Up Throw	Losing space	0.531	0.493	0.001	0.005	0.025	41
Grip Test	Broad Jump	Land to release	0.528	0.491	0.003	0.107	0.008	42

Pitch speed and individual predictors



